



Q.1. Answer the following short questions:

(6x5=30)

- Discuss the duality between moving average and auto-regressive process.
- Write a short note on stationarity of time series.
- Define stochastic process.
- Discuss about the invertibility of MA(q) process.
- Write down the objectives of time series analysis.
- Define Autocorrelation and Correlogram.

Answer the following questions.

(3x10=30)

Q.2. Given the following time series, compute the first two partial autocorrelations.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
X_t	13	8	15	4	4	12	11	7	14	12

Q.3. Derive the properties of AR(p) process.

Q.4. Given the following MA(1) processes

Model-1 : $X_t = Z_t + \beta Z_{t-1}$

Model-2 : $X_t = Z_t + \frac{1}{\beta} Z_{t-1}$

Where $\{Z_t\}$ is a purely random process with zero mean and finite variance σ^2

Find the autocorrelation functions of Model-1 and Model-2. Also comment on the invertibility condition of MA process by discussing the autocorrelation functions of both models.